

■ WASSCE

2026 Midnight Mock

CORE MATHEMATICS

ORDER: Paper 2 (Essay) comes first in this booklet, then Paper 1 (Objectives)

Paper 2: Section A: 5 compulsory $\times$ 8 marks = 40 | Section B: answer 5 of 8 $\times$ 12 marks = 60

Paper 1: 50 objective questions — ALL compulsory · 50 marks · 1h 30min

No: No construction questions. No calculus. High-probability topics only.

Answers: Full model answers and marking scheme at the end of this booklet.

Last-Night Strategy — Read Before You Start

- In Paper 2, do Section A completely first — all 5 are compulsory and straightforward.
- In Section B, read ALL 8 questions, then choose your best 5. Do not rush this choice.
- Word problems: underline key numbers, draw a diagram, THEN write equations.
- Ogive: plot UPPER CLASS BOUNDARY on the x-axis — never the midpoint.
- Show every step — method marks keep you scoring even when final answers are wrong.

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BEWARE OF EXAM SCAMMERS!

No one has WASSCE papers before the exam. Do NOT pay anyone. Your preparation is your power.

PAPER 2 — ESSAY

100 marks · 2 hours 30 minutes · No construction · No calculus

*All working must be shown clearly — method marks are awarded step by step.**For graph questions use graph paper. Label all axes and use the stated scale.**Non-programmable calculators are permitted. Always write units in final answers.***SECTION A — Answer ALL FIVE questions · 8 marks each****Question 1****[8 marks]****(a)**

A binary operation $\#$ is defined on the set $W = \{1, 2, 3, 4, 5, 6\}$ by

$p \# q = (p + q - 1) \bmod 6$, where a result of 0 is replaced by 6.

(i) Construct the operation table for $\#$ on W .

(ii) Find the value of k such that $4 \# k = 2$.

(iii) State one property of $\#$ that can be observed from the table. Justify your answer. **[4 marks]**

(b)

Abena deposited GHS 8,000 in a savings account at a compound interest rate of 12.5% per annum.

(i) Calculate the interest earned in the first year.

(ii) Find the total amount in the account after 2 years.

(iii) How much MORE interest does compound interest earn over 2 years compared to simple interest at the same rate? **[4 marks]**

Question 2**[8 marks]****(a)**

Solve the simultaneous equations:

$$2^{x+y} = 16 \text{ and } 3^x \times 9^y = 243$$

Hence find the value of $x - y$. **[4 marks]**

(b)

A cocoa farmer harvested 50 bags in the first year. Each year, his yield increases by 20% of the previous year's yield.

(i) Write down the yields for years 1, 2 and 3.

(ii) Find the yield in year 6, correct to the nearest whole number.

(iii) Calculate the total yield over 6 years, correct to the nearest whole number. **[4 marks]**

Question 3**[8 marks]****(a)**

(i) Given that $v^2 = u^2 + 2as$, make **a** the subject of the formula.

(ii) Find the value of a when $v = 10$, $u = 4$ and $s = 7$. **[4 marks]**

(b)

The sum of Kweku's age and his daughter's age is 66 years. Six years ago, Kweku was five times as old as his daughter.

(i) Write two equations to represent this information.

(ii) Solve the equations to find their current ages.

(iii) In how many years will the sum of their ages be 90? **[4 marks]**

Question 4

[8 marks]

(a)

PA and PB are tangents drawn from an external point P to a circle with centre O. PA = 8 cm and OP = 10 cm.

(i) Calculate the radius of the circle.

(ii) Find angle OPA, correct to the nearest degree.

(iii) Find angle AOB.

(iv) Find angle APB. **[4 marks]**

(b)

A child's toy consists of a solid cone placed exactly on top of a solid hemisphere. Both the cone and hemisphere have radius 3.5 cm. The height of the cone is 12 cm.

(i) Find the slant height of the cone.

(ii) Calculate the total volume of the toy.

(iii) Find the total surface area of the toy.

[Take $\pi = 22/7$] **[4 marks]**

Question 5

[8 marks]

(a)

Two straight lines have equations $3x - y = 7$ and $x + 2y = 7$.

(i) Find the coordinates of their point of intersection, P.

(ii) Find the equation of the straight line through P and Q(5, 6).

(iii) Where does this line through P and Q cross the x-axis? **[4 marks]**

(b)

Five cards are numbered 1, 2, 3, 4 and 5. Two cards are drawn at random, one after the other, without replacement.

Find the probability that:

(i) both cards show odd numbers;

(ii) the sum of the two numbers is greater than 6;

(iii) the product of the two numbers is even. **[4 marks]**

SECTION B — Answer any FIVE questions · 12 marks each

Question 6 [Quadratic Graph & Speed Word Problem]

[12 marks]

(a)

Copy and complete the table for $y = x^2 - 3x - 4$ for $-2 \leq x \leq 5$:

x	-2	-1	0	1	2	3	4	5
y		0	-4		-6		0	

(ii) Using a scale of 2 cm to 1 unit on the x-axis and 2 cm to 2 units on the y-axis, draw the graph of $y = x^2 - 3x - 4$.

(iii) From your graph, state: (α) the roots of $x^2 - 3x - 4 = 0$; (β) the minimum value of y and the value of x at which it occurs; (γ) the range of values of x for which $y < 0$. **[7 marks]**

(b)

A commercial bus travels 240 km from one town to another. On the return journey, the speed is reduced by 20 km/h and the journey takes 2 hours longer.

(i) If the outward speed is v km/h, write expressions for the time of each journey.

(ii) Form an equation in v , show that it simplifies to $v^2 - 20v - 2400 = 0$, and solve it.

(iii) Find the total time for the round trip. **[5 marks]**

Question 7 [Statistics — Ogive & Mean]

[12 marks]

(a)

The marks scored by 120 candidates in a test are shown below:

Mark	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
Freq	5	10	20	35	y	15	7	3

(i) Find the value of y .

(ii) Construct a cumulative frequency table using the UPPER CLASS BOUNDARY for the x-axis.

(iii) Draw a cumulative frequency curve (ogive). Scale: 2 cm to 10 marks; 2 cm to 10 candidates.

(iv) Use your ogive to estimate: (α) the median mark; (β) the interquartile range; (γ) the pass mark if 40% of candidates pass. **[6 marks]**

(b)

Using the same data: (i) calculate the mean mark; (ii) state the modal class;

(iii) find the probability that a candidate chosen at random scored 60 marks or more. **[6 marks]**

Question 8 [Bearings]

[12 marks]

(a)

Three vessels depart from port P. Vessel A sails 120 km on a bearing of 070° to reach an oil platform at Q. Vessel B then sails 90 km on a bearing of 160° from Q to reach a second platform at R.

- (i) Draw a clearly labelled diagram of the positions of P, Q and R.
- (ii) Show that angle $PQR = 90^\circ$.
- (iii) Calculate the distance PR. **[6 marks]**

(b)

- (i) Find the bearing of R from P, correct to the nearest degree.
- (ii) Find the bearing of P from R.

[Use: $\tan 36.87^\circ = 0.75$, $\sin 36.87^\circ = 0.6$, $\cos 36.87^\circ = 0.8$] **[6 marks]**

Question 9 [Linear Inequalities & Partial Variation]

[12 marks]

(a)

On graph paper, using a scale of 2 cm to 1 unit on both axes, draw the lines:

$$x = 0, y = 0, x + y = 6, x + 2y = 10.$$

- (i) Label the region R which satisfies all four inequalities simultaneously: $x \geq 0$, $y \geq 0$, $x + y \leq 6$ and $x + 2y \leq 10$.
- (ii) Write down the coordinates of all vertices of region R.
- (iii) Find the maximum value of $3x + 2y$ within region R, and state where it occurs. **[6 marks]**

(b)

The cost C (in GHS) of producing n items in a small factory is partly fixed and partly varies directly as n.

When $n = 200$, $C = \text{GHS } 1,500$. When $n = 600$, $C = \text{GHS } 3,500$.

- (i) Find an expression for C in terms of n.
- (ii) Find the cost of producing 350 items.
- (iii) Each item is sold for GHS 12. Find the profit per item when 500 items are produced. **[6 marks]**

Question 10 [Probability & Algebraic Fractions]

[12 marks]

(a)

A bag contains 5 blue, 4 red and 3 white sweets, all identical in size. Two sweets are drawn at random, one after the other, without replacement.

- (i) Draw a tree diagram to show all possible colour combinations for the two draws.
- (ii) Find the probability that: (α) both sweets are the same colour; (β) at least one sweet is blue; (γ) the first sweet is blue and the second is red; (δ) the two sweets are different colours. **[6 marks]**

(b)

- (i) Simplify completely: $(x^2 + 2x)/(x^2 + 5x + 6) \times (x + 3)/(x^2 - 4)$.

(ii) Solve: $2/(x - 1) + 3/(x + 2) = 2$. [6 marks]

Question 11 [Functions & AP Savings Problem]

[12 marks]

(a)

Two functions are defined as $f(x) = 2x + 3$ and $g(x) = x^2 - 1$.

(i) Find $fg(2)$.

(ii) Find $gf(x)$ and simplify your answer.

(iii) Find $f^{-1}(x)$ and hence solve $f^{-1}(x) = g(3)$.

(iv) Solve the equation $fg(x) = 0$. [6 marks]

(b)

Ama wants to save money to buy a laptop. In month 1 she saves GHS 300, and each subsequent month she saves GHS 50 more than the previous month.

(i) How much does she save in month 5?

(ii) In which month does she first save more than GHS 550 in a single month?

(iii) Show that her total savings after 10 months is exactly GHS 5,250.

(iv) She deposits GHS 5,250 in a bank at 8% simple interest per annum. After how many years will she have GHS 6,090? [6 marks]

Question 12 [Depreciation & Square Pyramid]

[12 marks]

[Take $\pi = 22/7$ where needed]

(a)

Kwame bought a car for GHS 80,000. The car depreciates at 15% in the first year and 20% per annum for each subsequent year.

(i) Find the value of the car after 3 years.

(ii) Calculate the total percentage depreciation over the 3 years.

At the same time, Kwame also invested GHS 20,000 in a savings account at 10% compound interest per annum.

(iii) Find the value of the investment after 3 years.

(iv) How much interest did the investment earn over 3 years? [6 marks]

(b)

A square-based pyramid has a base of side 12 cm and a perpendicular height of 8 cm.

(i) Find the slant height (from the apex to the midpoint of a base edge).

(ii) Calculate the volume of the pyramid.

(iii) Find the total surface area of the pyramid.

(iv) Find the length of the slant edge (from the apex to a corner of the base). [6 marks]

Question 13 [Statistics & Indices/Logarithms]

[12 marks]

(a)

The table shows the marks scored out of 5 by 30 students in a weekly quiz:

Mark (x)	0	1	2	3	4	5
Frequency (f)	1	4	10	8	5	2

- (i) Calculate the mean mark.
- (ii) Calculate the standard deviation, correct to 1 decimal place.
- (iii) Find the probability that a student chosen at random scored above the mean. **[6 marks]**

(b)

Solve each of the following equations:

- (i) $4^{(x+1)} = 2^{(3x-1)}$
- (ii) $3^{(2x)} + 3^x - 2 = 0$ [Hint: let $u = 3^x$]
- (iii) $\log_{\blacksquare}(x+2) + \log_{\blacksquare}(x-2) = \log_{\blacksquare} 12$
- (iv) $\log_{\blacksquare}(x+3) + \log_{\blacksquare}(x-1) = 5$ **[6 marks]**

PAPER 1 — OBJECTIVE TEST

Answer ALL 50 questions · Circle the letter of the correct answer · 50 marks · 1h 30min

1. Which of the following is equal to 1011001_2 in base 10?
 - A. 87
 - B. 88
 - C. 89
 - D. 90
2. Simplify: $\sqrt{75} + 2\sqrt{48} - \sqrt{27}$.
 - A. $10\sqrt{3}$
 - B. $12\sqrt{3}$
 - C. $14\sqrt{3}$
 - D. $16\sqrt{3}$
3. Express 0.000506 in standard form.
 - A. 5.06×10^{-3}
 - B. 5.06×10^{-4}
 - C. 5.06×10^{-5}
 - D. 50.6×10^{-4}
4. A binary operation $\oplus$ is defined by $x \oplus y = x + y - xy$. Find $3 \oplus 4$.
 - A. -5
 - B. -1
 - C. 1
 - D. 5
5. If $P = \{\text{prime numbers less than 12}\}$ and $Q = \{\text{odd numbers less than 12}\}$, find $n(P \cap Q)$.
 - A. 3
 - B. 4
 - C. 5
 - D. 6
6. Kofi borrowed GHS 15,000 at 8% simple interest per annum. After how many years does he owe GHS 18,600?
 - A. 2
 - B. 3
 - C. 4
 - D. 5
7. GHS 10,000 is invested at 10% per annum compound interest. Find the value after 2 years.
 - A. GHS 11,000
 - B. GHS 12,000
 - C. GHS 12,100
 - D. GHS 12,200
8. A car worth GHS 50,000 depreciates at 20% per annum. Find its value after 2 years.
 - A. GHS 30,000

- B. GHS 32,000
- C. GHS 34,000
- D. GHS 36,000

9. Solve: $5^{(2x-1)} = 25$.

- A. $x = 1$
- B. $x = 1.5$
- C. $x = 2$
- D. $x = 3$

10. Evaluate: $\log_2 32 - \log_2 4$.

- A. 2
- B. 3
- C. 4
- D. 8

11. The quadratic equation $2x^2 - 7x + 3 = 0$ has roots:

- A. $x = 3$ and $x = 1/2$
- B. $x = -3$ and $x = -1/2$
- C. $x = 3$ and $x = -1/2$
- D. $x = 1/2$ and $x = 1$

12. In an AP, $a = 5$ and $d = 4$. Which term equals 41?

- A. 8th
- B. 9th
- C. 10th
- D. 11th

13. y varies inversely as x^2 . When $x = 3$, $y = 4$. Find y when $x = 2$.

- A. 6
- B. 7
- C. 8
- D. 9

14. Find the gradient of the perpendicular bisector of the line joining (1, 3) and (5, 1).

- A. -2
- B. $-1/2$
- C. $1/2$
- D. 2

15. A rotation of 90° anticlockwise about the origin maps (3, -2) to:

- A. (-2, -3)
- B. (2, 3)
- C. (-3, 2)
- D. (2, -3)

16. A box contains 6 identical balls: 2 red, 3 blue and 1 yellow. $P(\text{not red}) =$

- A. $1/3$
- B. $2/3$
- C. $1/6$
- D. $5/6$

17. ABCD is a cyclic quadrilateral. Angle A = 105° . Find angle C.
- A. 75°
 - B. 85°
 - C. 95°
 - D. 105°
18. The arc length of a sector with radius 14 cm and angle 90° is $[\pi = 22/7]$:
- A. 11 cm
 - B. 22 cm
 - C. 33 cm
 - D. 44 cm
19. Find the total surface area of a sphere of radius 7 cm. $[\pi = 22/7]$
- A. 308 cm^2
 - B. 616 cm^2
 - C. 924 cm^2
 - D. 1232 cm^2
20. Given $a = (3, -4)$ and $b = (-1, 2)$, find $2a - b$.
- A. $(5, -6)$
 - B. $(7, -10)$
 - C. $(5, -10)$
 - D. $(7, -6)$
21. Find the range of the data: 12, 18, 7, 24, 15, 9, 21.
- A. 14
 - B. 15
 - C. 17
 - D. 20
22. Solve: $5 - 2x < x + 2$.
- A. $x > 1$
 - B. $x < 1$
 - C. $x > 3$
 - D. $x < 3$
23. Simplify: $(a^2 - b^2)/(a + b)$.
- A. $a - b$
 - B. $a + b$
 - C. $a^2 + b$
 - D. a^2b
24. Solve: $2/(x+1) = 3/(x+3)$.
- A. $x = 2$
 - B. $x = 3$
 - C. $x = 4$
 - D. $x = 5$
25. The sum of interior angles of an octagon is:
- A. 900°

- B. 1080°
- C. 1260°
- D. 1440°

26. Find the value of x if $3^x = 81$.

- A. 2
- B. 3
- C. 4
- D. 5

27. The LCM of 12, 18 and 24 is:

- A. 36
- B. 48
- C. 72
- D. 144

28. A man walks 15 km due South then 8 km due East. How far is he from his starting point?

- A. 7 km
- B. 17 km
- C. 19 km
- D. 23 km

29. What is the bearing of North-East?

- A. 045°
- B. 090°
- C. 135°
- D. 315°

30. If $A \subset B$ and $n(B) = 12$, $n(A) = 7$, find $n(B - A)$.

- A. 5
- B. 7
- C. 12
- D. 19

31. Evaluate: $(\frac{4}{9})^{-1/2}$.

- A. $\frac{2}{3}$
- B. $\frac{3}{2}$
- C. $\frac{4}{9}$
- D. $\frac{9}{4}$

32. The 4th term of a GP is 24 and the 7th term is 192. Find the common ratio.

- A. 1
- B. 2
- C. 3
- D. 4

33. Find the area of a sector with radius 21 cm and angle 60° . [$\pi = \frac{22}{7}$]

- A. 231 cm^2
- B. 242 cm^2
- C. 253 cm^2
- D. 264 cm^2

34. Solve: $4 \log x = \log 81$.
- A. $x = 2$
 - B. $x = 3$
 - C. $x = 4$
 - D. $x = 9$
35. A die is rolled twice. $P(5 \text{ on first and } 5 \text{ on second}) =$
- A. $1/6$
 - B. $1/12$
 - C. $1/36$
 - D. $5/36$
36. What is 18% of GHS 2,500?
- A. GHS 350
 - B. GHS 400
 - C. GHS 425
 - D. GHS 450
37. The mean of 5, 8, x , 12, 10 is 9. Find x .
- A. 8
 - B. 9
 - C. 10
 - D. 11
38. A line has equation $y = -(1/3)x + 5$. Its x -intercept is:
- A. (5, 0)
 - B. (15, 0)
 - C. (0, 5)
 - D. (-15, 0)
39. If $g(x) = x^2 - 3$, find $g(g(2))$.
- A. 1
 - B. 2
 - C. 3
 - D. 4
40. Two cards are drawn from a pack of 52. $P(\text{both aces}) =$
- A. $1/221$
 - B. $1/169$
 - C. $2/52$
 - D. $1/26$
41. A train 150 m long passes a pole in 15 seconds. Find its speed in km/h.
- A. 32 km/h
 - B. 36 km/h
 - C. 40 km/h
 - D. 45 km/h
42. The image of (2, 5) after a translation by vector $(-3, 1)$ is:
- A. (-1, 6)

- B. (5, 4)
- C. (-1, 4)
- D. (5, 6)

43. A cone has base radius 7 cm and slant height 25 cm. Its curved surface area is [$\pi = 22/7$]:

- A. 450 cm²
- B. 550 cm²
- C. 700 cm²
- D. 850 cm²

44. An angle subtended at the centre by a chord is 80°. What is the angle in the alternate segment?

- A. 40°
- B. 50°
- C. 80°
- D. 100°

45. Find the sum of the first 8 terms of the AP: 3, 7, 11, 15, ...

- A. 96
- B. 100
- C. 104
- D. 108

46. A bag has 4 green and 6 orange sweets. One is taken and not replaced; then another is taken. P(both orange) =

- A. 1/3
- B. 3/9
- C. 1/3
- D. 6/25

47. Rationalise: $6/(2 - \sqrt{2})$.

- A. $3(2+\sqrt{2})$
- B. $6(2+\sqrt{2})/2$
- C. $2(2+\sqrt{2})$
- D. $3+3\sqrt{2}$

48. A pole 6 m tall casts a shadow 4 m long. Find the angle of elevation of the sun.

- A. 42°
- B. 48°
- C. 56°
- D. 62°

49. What is the equation of the line through (0, 4) perpendicular to $2x - y = 3$?

- A. $y = 2x + 4$
- B. $y = -(1/2)x + 4$
- C. $y = (1/2)x + 4$
- D. $y = -2x + 4$

50. A GP has first term 3 and sum to infinity 12. Find the common ratio.

- A. 1/4
- B. 1/3
- C. 1/2

D. $\frac{3}{4}$

ANSWERS — PAPER 2 (ESSAY)

Attempt ALL questions BEFORE checking · M = method · A = accuracy

A correct answer without working may score zero. Show every step.

SECTION A — MODEL ANSWERS

Question 1

Q1(a): Binary Operation Table [4 marks]

$p \# q = (p + q - 1) \bmod 6$, where 0 is replaced by 6.

Table (rows = p, columns = q):

- # | 1 2 3 4 5 6
- 1 | 1 2 3 4 5 6
- 2 | 2 3 4 5 6 1
- 3 | 3 4 5 6 1 2
- 4 | 4 5 6 1 2 3
- 5 | 5 6 1 2 3 4
- 6 | 6 1 2 3 4 5

(ii) Find k: $4 \# k = 2$. $(4+k-1) \bmod 6 = 2 \rightarrow (3+k) \bmod 6 = 2 \rightarrow k = 5$.

Verify: $(4+5-1) \bmod 6 = 8 \bmod 6 = 2 \checkmark$

(iii) Property: the operation # is COMMUTATIVE ($p \# q = q \# p$ for all $p, q \in W$), since the table is symmetric about the main diagonal.

Also: the operation is CLOSED, since every result is in $W = \{1,2,3,4,5,6\}$.

Marking: M1 correct method. A1 at least 5 correct rows. A1 $k=5$. A1 correct property with justification.

Q1(b): Compound Interest [4 marks]

$P = \text{GHS } 8,000$. Rate = 12.5% per annum.

(i) Interest in year 1 = $12.5\% \times 8,000 = \text{GHS } 1,000$.

Amount after year 1 = GHS 9,000.

(ii) Interest in year 2 = $12.5\% \times 9,000 = \text{GHS } 1,125$.

Total amount after 2 years = $9,000 + 1,125 = \text{GHS } 10,125$.

(iii) Simple interest over 2 years = $8,000 \times 12.5\% \times 2 = \text{GHS } 2,000$.

Compound interest earned = $10,125 - 8,000 = \text{GHS } 2,125$.

Extra from compounding = $2,125 - 2,000 = \text{GHS } 125$.

Marking: A1 year 1 interest = GHS 1,000. A1 total after 2 years = GHS 10,125. A1 SI = GHS 2,000. A1 extra = GHS 125.

Question 2

Q2(a): Simultaneous Indices Equations [4 marks]

$$2^{x+y} = 16 = 2^4 \rightarrow x + y = 4 \dots \text{(equation 1)}$$

$$3^x \times 9^y = 243. \text{ Since } 9 = 3^2 \text{ and } 243 = 3^5: 3^x \times 3^{2y} = 3^5 \rightarrow x + 2y = 5 \dots \text{(equation 2)}$$

Subtract equation (1) from equation (2): $y = 1$.

Substitute into equation (1): $x = 4 - 1 = 3$.

$$\text{Verify: } 2^{3+1} = 2^4 = 16 \checkmark. 3^3 \times 9^1 = 27 \times 9 = 243 \checkmark.$$

$$x - y = 3 - 1 = 2.$$

Marking: M1 converting to index form. A1 two linear equations. M1 solving. A1 $x=3$, $y=1$. A1 $x-y=2$.

Q2(b): Geometric Progression — Cocoa Farm [4 marks]

First term $a = 50$ bags. Common ratio $r = 1.20$ (20% increase each year).

$$(i) \text{ Year 1: } 50. \text{ Year 2: } 50 \times 1.2 = 60. \text{ Year 3: } 60 \times 1.2 = 72 \text{ bags.}$$

$$(ii) \text{ Year 6: } T_n = 50 \times (1.2)^5 = 50 \times 2.48832 = 124.4 \approx 124 \text{ bags.}$$

$$(iii) S_n = a(r^n - 1)/(r - 1) = 50 \times (1.2^6 - 1) / (1.2 - 1)$$

$$= 50 \times (2.985984 - 1) / 0.2 = 50 \times 1.985984 / 0.2$$

$$= 50 \times 9.92992 = 496.5 \approx 497 \text{ bags.}$$

Marking: A1 three correct terms. A1 $T_n \approx 124$ bags. M1 correct sum formula. A1 $S_n \approx 497$ bags.

Question 3**Q3(a): Formula Manipulation — Make a the Subject [4 marks]**

$$\text{Given: } v^2 = u^2 + 2as.$$

$$\text{Step 1: } v^2 - u^2 = 2as.$$

$$\text{Step 2: } a = (v^2 - u^2) / (2s).$$

Finding a when $v = 10$, $u = 4$, $s = 7$:

$$a = (10^2 - 4^2) / (2 \times 7) = (100 - 16) / 14 = 84 / 14 = 6 \text{ m/s}^2.$$

Marking: M1 isolating a s. A1 $a = (v^2 - u^2)/2s$. M1 substitution. A1 $a = 6 \text{ m/s}^2$.

Q3(b): Age Word Problem [4 marks]

Let Kweku's age = K and daughter's age = D (both current ages).

$$\text{Equation 1 (sum of ages): } K + D = 66.$$

$$\text{Equation 2 (six years ago): } K - 6 = 5(D - 6) \rightarrow K = 5D - 24.$$

$$\text{Substitute into Equation 1: } (5D - 24) + D = 66 \rightarrow 6D = 90 \rightarrow D = 15 \text{ years.}$$

$$K = 66 - 15 = 51 \text{ years.}$$

$$\text{Verify: 6 years ago: } D = 9, K = 45. 45 = 5 \times 9 \checkmark.$$

$$(iii) \text{ Sum now} = 66. \text{ Each year, sum increases by 2.}$$

$$\text{After } n \text{ years: } 66 + 2n = 90 \rightarrow 2n = 24 \rightarrow n = 12 \text{ years.}$$

Marking: M1 setting up two equations. A1 $D = 15$. A1 $K = 51$. A1 12 years.

Question 4

Q4(a): Tangents from External Point [4 marks]

In right-angled triangle OAP: $OA \perp PA$ (radius to tangent is perpendicular).

$$OA^2 + PA^2 = OP^2 \rightarrow OA^2 + 8^2 = 10^2 \rightarrow OA^2 = 100 - 64 = 36 \rightarrow OA = 6 \text{ cm.}$$

$$(ii) \tan(\angle OPA) = OA/PA = 6/8 = 0.75 \rightarrow \angle OPA = \tan^{-1}(0.75) \approx 37^\circ.$$

$$(iii) \angle AOP = 90^\circ - 37^\circ = 53^\circ. \text{ By symmetry } \angle BOP = 53^\circ.$$

$$\angle AOB = 2 \times 53^\circ = 106^\circ.$$

$$(iv) \angle APB = 2 \times \angle OPA = 2 \times 37^\circ = 74^\circ.$$

[Alternatively: $\angle APB + \angle AOB = 180^\circ$ (angle between tangents + angle at centre = 180°)]

Marking: M1 Pythagoras. A1 $r=6\text{cm}$. A1 $\angle OPA=37^\circ$. A1 $\angle AOB=106^\circ$. A1 $\angle APB=74^\circ$.

Q4(b): Cone + Hemisphere Toy [4 marks]

Radius $r = 3.5 \text{ cm}$. Cone height $h = 12 \text{ cm}$. [$\pi = 22/7$]

$$(i) \text{ Slant height } l = \sqrt{(h^2 + r^2)} = \sqrt{(144 + 12.25)} = \sqrt{156.25} = 12.5 \text{ cm.}$$

$$(ii) \text{ Volume of cone} = (1/3)\pi r^2 h = (1/3) \times (22/7) \times 12.25 \times 12 = 154 \text{ cm}^3.$$

$$\text{Volume of hemisphere} = (2/3)\pi r^3 = (2/3) \times (22/7) \times 42.875$$

$$= (2 \times 22 \times 42.875) / (3 \times 7) = 1887.5/21 \approx 89.88 \text{ cm}^3.$$

$$\text{Total volume} = 154 + 89.88 = 243.88 \approx 243.9 \text{ cm}^3.$$

$$(iii) \text{ Curved surface of cone} = \pi r l = (22/7) \times 3.5 \times 12.5 = 137.5 \text{ cm}^2.$$

$$\text{Curved surface of hemisphere} = 2\pi r^2 = 2 \times (22/7) \times 12.25 = 77 \text{ cm}^2.$$

$$\text{Total surface area} = 137.5 + 77 = 214.5 \text{ cm}^2.$$

Marking: A1 $l=12.5\text{cm}$. M1 each volume formula. A1 total $V \approx 243.9\text{cm}^3$. M1 SA method. A1 $SA=214.5\text{cm}^2$.

Question 5**Q5(a): Coordinate Geometry — Intersection and Line [4 marks]**

Equations: $3x - y = 7 \dots (1)$ and $x + 2y = 7 \dots (2)$.

From (1): $y = 3x - 7$. Substitute into (2): $x + 2(3x - 7) = 7 \rightarrow 7x = 21 \rightarrow x = 3$.

$y = 3(3) - 7 = 2$. Point of intersection $P = (3, 2)$.

(ii) Gradient of line through $P(3, 2)$ and $Q(5, 6)$:

$$m = (6 - 2) / (5 - 3) = 4/2 = 2.$$

$$\text{Equation: } y - 2 = 2(x - 3) \rightarrow y = 2x - 4 \rightarrow 2x - y = 4.$$

(iii) Set $y = 0$: $2x = 4 \rightarrow x = 2$. Line crosses x-axis at $(2, 0)$.

Marking: M1 substitution. A1 $P=(3,2)$. M1 gradient. A1 equation $y=2x-4$. A1 $(2,0)$.

Q5(b): Probability — Number Cards [4 marks]

Cards: $\{1, 2, 3, 4, 5\}$. Two drawn without replacement.

Total ordered pairs = $5 \times 4 = 20$.

$$(i) \text{ Odd numbers in set: } \{1, 3, 5\}. P(\text{both odd}) = (3/5) \times (2/4) = 6/20 = 3/10.$$

$$(ii) \text{ Pairs with sum } > 6: (2,5),(5,2),(3,4),(4,3),(3,5),(5,3),(4,5),(5,4) = 8 \text{ pairs.}$$

$$P(\text{sum} > 6) = 8/20 = 2/5.$$

(iii) Product is even when at least one card is even. Even cards: $\{2, 4\}$.

$$P(\text{product even}) = 1 - P(\text{both odd}) = 1 - 3/10 = 7/10.$$

Marking: A1 $P(\text{both odd}) = 3/10$. M1 listing pairs for $\text{sum} > 6$. A1 $P(\text{sum} > 6) = 2/5$. A1 $P(\text{even product}) = 7/10$.

SECTION B — MODEL ANSWERS

Q6(a): Quadratic Graph $y = x^2 - 3x - 4$ [7 marks]

Complete the table for $-2 \leq x \leq 5$:

- $x=-2$: $4+6-4=6$. $x=-1$: $1+3-4=0$. $x=0$: -4 . $x=1$: $1-3-4=-6$.
- $x=2$: $4-6-4=-6$. $x=3$: $9-9-4=-4$. $x=4$: $16-12-4=0$. $x=5$: $25-15-4=6$.

Scale: 2 cm to 1 unit (x-axis); 2 cm to 2 units (y-axis). Plot smooth U-shaped parabola.

Roots ($y=0$): $x = -1$ and $x = 4$. [Verify: $(x+1)(x-4)=0$ ✓]

Minimum: axis of symmetry $x = -(-3)/2 = 1.5$. $y_{\min} = 2.25 - 4.5 - 4 = -6.25$.

Minimum point: (1.5, -6.25).

Range where $y < 0$: from graph, $-1 < x < 4$.

Marking: M1 correct table. M2 correct plotted points. A1 smooth curve. A1 roots. A1 minimum. A1 range.

Q6(b): Speed-Distance Word Problem [5 marks]

Distance = 240 km. Let outward speed = v km/h. Return speed = $(v-20)$ km/h.

(i) Outward time = $240/v$ hours. Return time = $240/(v-20)$ hours.

(ii) Return takes 2 hours longer: $240/(v-20) - 240/v = 2$.

Multiply through by $v(v-20)$:

$$240v - 240(v-20) = 2v(v-20)$$

$$240v - 240v + 4800 = 2v^2 - 40v$$

$$4800 = 2v^2 - 40v \rightarrow v^2 - 20v - 2400 = 0.$$

Factorise: $(v - 60)(v + 40) = 0 \rightarrow v = 60$ (reject $v = -40$).

Outward speed = 60 km/h. Return speed = 40 km/h.

(iii) Total time = $240/60 + 240/40 = 4 + 6 = 10$ hours.

Marking: M1 time expressions. M1 correct equation. M1 form quadratic. A1 $v=60$ km/h. A1 total=10h.

Q7(a): Ogive — 120 Candidates [6 marks]

Find y : $5+10+20+35+y+15+7+3=120 \rightarrow 95+y=120 \rightarrow y=25$.

Cumulative frequency table — ALWAYS plot against UPPER CLASS BOUNDARY:

- $19.5 \rightarrow 5$, $29.5 \rightarrow 15$, $39.5 \rightarrow 35$, $49.5 \rightarrow 70$, $59.5 \rightarrow 95$, $69.5 \rightarrow 110$, $79.5 \rightarrow 117$, $89.5 \rightarrow 120$.

Draw ogive: begin at (9.5, 0). Plot 8 points. Join with smooth S-shaped curve.

Scale: 2 cm = 10 marks (x-axis), 2 cm = 10 candidates (y-axis).

From ogive: Median at $CF=60 \rightarrow \approx 48$ marks.

Q1 at $CF=30 \rightarrow \approx 38$ marks. Q3 at $CF=90 \rightarrow \approx 62$ marks. $IQR = 62 - 38 = 24$.

Pass mark if 40% pass (72 pass, 48 fail): read at $CF=48 \rightarrow \approx 46$ marks.

Marking: A1 $y=25$. M1 correct UCB. A1 CF table. M2 correct ogive. A1 each estimate from graph.

Q7(b): Mean from Grouped Data [6 marks]

Midpoints: 14.5, 24.5, 34.5, 44.5, 54.5, 64.5, 74.5, 84.5.

Frequencies: 5, 10, 20, 35, 25, 15, 7, 3. Total = 120.

$$\begin{aligned}\Sigma fx &= 5(14.5) + 10(24.5) + 20(34.5) + 35(44.5) + 25(54.5) + 15(64.5) + 7(74.5) + 3(84.5) \\ &= 72.5 + 245 + 690 + 1557.5 + 1362.5 + 967.5 + 521.5 + 253.5 = 5670.\end{aligned}$$

Mean = $5670 / 120 = 47.25$ marks.

Modal class: 40–49 (highest frequency = 35).

P(score ≥ 60) from ogive: $120 - 95 = 25$ candidates. $P = 25/120 = 5/24$.

Marking: M1 midpoints. M1 fx column. A1 $\Sigma fx = 5670$. A1 mean = 47.25. A1 modal class. A1 probability.

Q8(a): Bearings — Three Vessels [6 marks]

From port P: vessel A sails 120 km on bearing 070° to platform Q.

From Q: vessel B sails 90 km on bearing 160° to platform R.

Step 1 — Find angle PQR:

Back-bearing from Q to P = $070^\circ + 180^\circ = 250^\circ$.

Bearing from Q to R = 160° .

Angle PQR = $250^\circ - 160^\circ = 90^\circ$.

Step 2 — Triangle PQR has a right angle at Q. Apply Pythagoras:

$$PR^2 = PQ^2 + QR^2 = 120^2 + 90^2 = 14,400 + 8,100 = 22,500.$$

$$PR = \sqrt{22,500} = 150 \text{ km}.$$

Step 3 — Find bearing of R from P:

$$\tan(\angle QPR) = QR/PQ = 90/120 = 0.75 \rightarrow \angle QPR = 36.87^\circ \approx 37^\circ.$$

$$\text{Bearing of R from P} = \text{bearing of Q from P} + \angle QPR = 070^\circ + 37^\circ = 107^\circ.$$

Marking: M1 diagram. M1 angle PQR = 90° . M1 Pythagoras. A1 PR = 150 km. M1 trig. A1 bearing = 107° .

Q8(b): Bearing of P from R [6 marks]

$$\text{Back-bearing of P from R} = \text{bearing of R from P} + 180^\circ = 107^\circ + 180^\circ = 287^\circ.$$

Full triangle solution summary:

PQ = 120 km (outward leg, bearing 070°).

QR = 90 km (second leg, bearing 160°).

Angle at Q = 90° (right angle — key insight).

PR = 150 km (hypotenuse).

Bearing of R from P = 107° .

Bearing of P from R = 287° .

Additional check using angle QPR = 37° and angle QRP = 53° :

$$\sin(\angle QRP) = QP/PR \dots \text{using sin rule: } \sin(QRP)/120 = \sin 90^\circ/150 \rightarrow \sin(QRP) = 0.8 \rightarrow QRP = 53^\circ. \checkmark$$

Marking: A1 back-bearing 287° . A1 verification. Full marks for complete labelled diagram with all values.

Q9(a): Linear Inequalities [6 marks]

Shade the region R satisfying: $x \geq 0$, $y \geq 0$, $x + y \leq 6$, $x + 2y \leq 10$.

Find vertices of the feasible region:

- Origin (0,0): satisfies all. ✓
- (6,0): $x+y=6$, $y=0$. ✓
- (2,4): intersection of $x+y=6$ and $x+2y=10$. Subtract: $y=4$, $x=2$. ✓
- (0,5): $x+2y=10$, $x=0$. ✓

Maximum value of $3x + 2y$ at each vertex:

(0,0): 0. (6,0): 18. (2,4): $6+8=14$. (0,5): 10.

Maximum value of $3x + 2y = 18$, occurring at the point (6, 0).

Minimum value = 0 at origin (0,0).

Marking: M1 correct graph with 4 boundary lines. M1 identify feasible region. A1 correct vertices. A1 max=18 at (6,0).

Q9(b): Partial Variation — Production Cost [6 marks]

C is partly fixed and partly varies directly as n: $C = a + bn$.

When $n = 200$, $C = 1,500$: $a + 200b = 1,500$... (1)

When $n = 600$, $C = 3,500$: $a + 600b = 3,500$... (2)

Subtract (1) from (2): $400b = 2,000 \rightarrow b = 5$.

From (1): $a = 1,500 - 1,000 = 500$.

Expression: $C = 500 + 5n$.

(i) When $n = 350$: $C = 500 + 1,750 = \text{GHS } 2,250$.

(ii) When $C = 3,000$: $3,000 = 500 + 5n \rightarrow n = 500$ items.

(iii) Revenue = $12n$. Profit = $12n - (500+5n) = 7n - 500$.

When $n = 500$: Profit = $3,500 - 500 = \text{GHS } 3,000$.

Profit per item = $3,000 \div 500 = \text{GHS } 6$.

Marking: M1 form $C=a+bn$. M1 two equations. A1 $b=5$, $a=500$. A1 C when $n=350$. A1 $n=500$. A1 profit/item=GHS6.

Q10(a): Probability — Sweets Without Replacement [6 marks]

Bag: 5 Blue (B), 4 Red (R), 3 White (W). Total = 12. Two drawn without replacement.

Total ways to choose 2 from 12 = $C(12,2) = 66$.

P(both same colour):

$P(BB) = C(5,2)/66 = 10/66$.

$P(RR) = C(4,2)/66 = 6/66$.

$P(WW) = C(3,2)/66 = 3/66$.

$P(\text{same}) = 19/66$.

$P(\text{at least one blue}) = 1 - P(\text{no blue}) = 1 - C(7,2)/66 = 1 - 21/66 = 45/66 = 15/22$.

$P(\text{first blue, second red}) = (5/12)(4/11) = 20/132 = 5/33$.

$P(\text{different colours}) = 1 - 19/66 = 47/66$.

Marking: M1 $C(12,2)=66$. A1 $P(\text{same})=19/66$. A1 $P(\geq 1 \text{ blue})=15/22$. A1 $P(1\text{st B, } 2\text{nd R})=5/33$. A1 $P(\text{diff})=47/66$.

Q10(b): Algebraic Fractions [6 marks]

(i) Simplify: $(x^2+2x)/(x^2+5x+6) \times (x+3)/(x^2-4)$.

Factorise each part:

$$x^2+2x = x(x+2), \quad x^2+5x+6 = (x+2)(x+3), \quad x^2-4 = (x+2)(x-2).$$

$$\text{Expression} = [x(x+2) / (x+2)(x+3)] \times [(x+3) / (x+2)(x-2)]$$

$$= [x/(x+3)] \times [(x+3)/(x+2)(x-2)]$$

$$= x / (x+2)(x-2) = x / (x^2-4).$$

(ii) Solve: $2/(x-1) + 3/(x+2) = 2$.

Multiply through by $(x-1)(x+2)$:

$$2(x+2) + 3(x-1) = 2(x-1)(x+2)$$

$$2x+4 + 3x-3 = 2(x^2+x-2)$$

$$5x+1 = 2x^2+2x-4$$

$$2x^2-3x-5 = 0.$$

Factorise: $(2x-5)(x+1) = 0 \rightarrow x = 5/2$ or $x = -1$.

Marking: M1 factorise each part. A1 $x/(x^2-4)$. M1 multiply by LCD. M1 form quadratic. A1 $x=5/2$. A1 $x=-1$.

Q11(a): Functions [6 marks]

$$f(x) = 2x + 3, \quad g(x) = x^2 - 1.$$

$$(i) fg(2) = f(g(2)) = f(4-1) = f(3) = 2(3)+3 = 9.$$

$$(ii) gf(x) = g(f(x)) = g(2x+3) = (2x+3)^2 - 1 = 4x^2+12x+9-1 = 4x^2+12x+8.$$

$$(iii) f^{-1}(x): \text{let } y = 2x+3 \rightarrow x = (y-3)/2 \rightarrow f^{-1}(x) = (x-3)/2.$$

$$\text{Solve } f^{-1}(x) = g(3): g(3) = 9-1 = 8.$$

$$(x-3)/2 = 8 \rightarrow x-3 = 16 \rightarrow x = 19.$$

$$(iv) \text{Solve } fg(x) = 0: 4x^2+12x+8 = 0 \rightarrow x^2+3x+2 = 0 \rightarrow (x+1)(x+2) = 0.$$

$$x = -1 \text{ or } x = -2.$$

Marking: A1 $fg(2)=9$. A1 $gf(x)=4x^2+12x+8$. M1 inverse method. A1 $f^{-1}(x)$. A1 $x=19$. A1 $x=-1, -2$.

Q11(b): AP Savings Problem [6 marks]

Ama saves GHS 300 in month 1 and increases savings by GHS 50 each month.

This is an AP with $a = 300$ and $d = 50$.

$$(i) \text{Savings in month 5: } T_5 = 300 + 4(50) = \text{GHS } 500.$$

(ii) Month in which she first saves more than GHS 550:

$$T_n = 300 + 50(n-1) > 550 \rightarrow 50(n-1) > 250 \rightarrow n > 6 \rightarrow n = 7.$$

$$\text{Month 7: } T_7 = 300 + 6(50) = \text{GHS } 600. \checkmark$$

$$(iii) \text{Total savings in 10 months: } S_{10} = 10/2 \times (2 \times 300 + 9 \times 50) = 5 \times (600+450) = 5 \times 1,050 = \text{GHS } 5,250.$$

(iv) GHS 5,250 in bank at 8% simple interest. Time to reach GHS 6,090:

$$\text{Interest needed} = 6,090 - 5,250 = \text{GHS } 840.$$

$$T = I \times 100 / (P \times R) = 840 \times 100 / (5,250 \times 8) = 84,000 / 42,000 = 2 \text{ years}.$$

Marking: A1 $T_5=\text{GHS}500$. A1 month 7. M1 S_{10} formula. A1 $S_{10}=\text{GHS}5,250$. M1 SI formula. A1 2 years.

Q12(a): Depreciation & Compound Interest [6 marks]

Car: GHS 80,000. Depreciates 15% year 1, 20% per year years 2 and 3.

Value after year 1 = $80,000 \times (1 - 0.15) = 80,000 \times 0.85 = \text{GHS } 68,000$.

Value after year 2 = $68,000 \times 0.80 = \text{GHS } 54,400$.

Value after year 3 = $54,400 \times 0.80 = \text{GHS } 43,520$.

% depreciation over 3 years = $(80,000 - 43,520) / 80,000 \times 100 = 36,480 / 80,000 \times 100 = 45.6\%$.

Investment: GHS 20,000 at 10% compound interest per annum for 3 years.

$A = 20,000 \times (1.10)^3 = 20,000 \times 1.331 = \text{GHS } 26,620$.

Interest earned = $26,620 - 20,000 = \text{GHS } 6,620$.

Marking: M1 each year depreciation. A1 GHS 43,520. A1 45.6%. M1 CI formula. A1 GHS 26,620. A1 GHS 6,620.

Q12(b): Square-Based Pyramid [6 marks]

Square base: side 12 cm. Perpendicular height: 8 cm.

(i) Slant height (from apex to midpoint of base edge):

$l = \sqrt{(h^2 + (\text{side}/2)^2)} = \sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10 \text{ cm}$.

(ii) Volume = $(1/3) \times \text{base area} \times \text{height} = (1/3) \times 12^2 \times 8 = (1/3) \times 144 \times 8 = 384 \text{ cm}^3$.

(iii) Lateral (curved) surface area = $4 \times (1/2 \times \text{base} \times \text{slant height})$

$= 4 \times (1/2 \times 12 \times 10) = 4 \times 60 = 240 \text{ cm}^2$.

Total surface area = lateral SA + base area = $240 + 144 = 384 \text{ cm}^2$.

(iv) Slant edge (apex to corner): half-diagonal of base = $\sqrt{(6^2 + 6^2)} = 6\sqrt{2}$.

Slant edge = $\sqrt{(8^2 + (6\sqrt{2})^2)} = \sqrt{(64 + 72)} = \sqrt{136} = 2\sqrt{34} \approx 11.66 \text{ cm}$.

Marking: A1 slant $h=10\text{cm}$. A1 $V=384\text{cm}^3$. M1 lateral SA method. A1 lateral $SA=240\text{cm}^2$. A1 total $SA=384\text{cm}^2$. A1 slant edge $=\sqrt{136}$.

Q13(a): Mean and Standard Deviation [6 marks]

Quiz marks out of 5 scored by 30 students. Marks: 0,1,2,3,4,5. Freq: 1,4,10,8,5,2. Total=30.

$\Sigma fx = 0(1) + 1(4) + 2(10) + 3(8) + 4(5) + 5(2) = 0 + 4 + 20 + 24 + 20 + 10 = 78$.

Mean = $78/30 = 2.6$ marks.

Standard deviation — build deviation table:

- Mark 0: $(0 - 2.6)^2 = 6.76$; $f=1$; $f(x - \bar{x})^2 = 6.76$
- Mark 1: $(1 - 2.6)^2 = 2.56$; $f=4$; $f(x - \bar{x})^2 = 10.24$
- Mark 2: $(2 - 2.6)^2 = 0.36$; $f=10$; $f(x - \bar{x})^2 = 3.60$
- Mark 3: $(3 - 2.6)^2 = 0.16$; $f=8$; $f(x - \bar{x})^2 = 1.28$
- Mark 4: $(4 - 2.6)^2 = 1.96$; $f=5$; $f(x - \bar{x})^2 = 9.80$
- Mark 5: $(5 - 2.6)^2 = 5.76$; $f=2$; $f(x - \bar{x})^2 = 11.52$

$\Sigma f(x - \bar{x})^2 = 6.76 + 10.24 + 3.60 + 1.28 + 9.80 + 11.52 = 43.20$.

Variance = $43.20/30 = 1.44$. Standard Deviation = $\sqrt{1.44} = 1.2$ marks.

Marking: M1 mean method. A1 mean=2.6. M2 correct deviation table. A1 variance=1.44. A1 SD=1.2.

Q13(b): Indices and Logarithm Equations [6 marks](i) Solve $4^{(x+1)} = 2^{(3x-1)}$:

$$2^{(2(x+1))} = 2^{(3x-1)} \rightarrow 2x+2 = 3x-1 \rightarrow x = 3.$$

Verify: $4^4 = 256 = 2^8$ ✓.(ii) Solve $3^{(2x)} + 3^x - 2 = 0$:

$$\text{Let } u = 3^x: u^2 + u - 2 = 0 \rightarrow (u+2)(u-1) = 0.$$

$$u = 1 \text{ (reject } u = -2 \text{ since } 3^x > 0). \text{ So } 3^x = 1 = 3^0 \rightarrow x = 0.$$

(iii) Solve $\log_{\blacksquare}(x+2) + \log_{\blacksquare}(x-2) = \log_{\blacksquare} 12$:

$$\log_{\blacksquare}[(x+2)(x-2)] = \log_{\blacksquare} 12 \rightarrow (x+2)(x-2) = 12.$$

$$x^2 - 4 = 12 \rightarrow x^2 = 16 \rightarrow x = 4 \text{ (reject } x = -4, \text{ makes } \log_{\blacksquare}(-2) \text{ undefined).}$$

Verify: $\log_{\blacksquare}(6) + \log_{\blacksquare}(2) = \log_{\blacksquare}(12)$ ✓.(iv) Solve $\log_{\blacksquare}(x+3) + \log_{\blacksquare}(x-1) = 5$:

$$\log_{\blacksquare}[(x+3)(x-1)] = 5 \rightarrow (x+3)(x-1) = 2^5 = 32.$$

$$x^2 + 2x - 3 = 32 \rightarrow x^2 + 2x - 35 = 0 \rightarrow (x+7)(x-5) = 0 \rightarrow x = 5.$$

[Reject $x = -7$ since $\log_{\blacksquare}(-4)$ is undefined].Verify: $\log_{\blacksquare}(8) + \log_{\blacksquare}(4) = 3 + 2 = 5$ ✓.**Marking:** A1 $x=3$. A1 $x=0$ (with substitution $u=3^x$). A1 $x=4$. A1 $x=5$. M1 method marks throughout.

ANSWERS — PAPER 1 (OBJECTIVES)

Study WHY each answer is correct — not just WHAT it is

1. C	2. B	3. B	4. A	5. B
6. B	7. C	8. B	9. B	10. B
11. A	12. C	13. D	14. D	15. B
16. B	17. A	18. B	19. B	20. B
21. C	22. A	23. A	24. B	25. B
26. C	27. C	28. B	29. A	30. A
31. B	32. B	33. A	34. B	35. C
36. D	37. C	38. B	39. A	40. A
41. B	42. A	43. B	44. A	45. C
46. A	47. A	48. C	49. B	50. D

Notes on key answers:

Q4: A — $3 \oplus 4 = 3 + 4 - 3 \times 4 = 7 - 12 = -5$.

Q11: A — $2x^2 - 7x + 3 = 0$. Factors: $(2x-1)(x-3) = 0 \rightarrow x = 1/2$ or $x = 3$.

Q31: B — $(4/9)^{-1/2} = (9/4)^{1/2} = 3/2$.

Q40: A — $P(\text{both aces}) = (4/52)(3/51) = 12/2652 = 1/221$.

Q47: A — $6/(2-\sqrt{2}) \times (2+\sqrt{2})/(2+\sqrt{2}) = 6(2+\sqrt{2})/(4-2) = 6(2+\sqrt{2})/2 = 3(2+\sqrt{2})$.

Q50: D — $S_{\infty} = a/(1-r) = 12$. $a = 3$. $3/(1-r) = 12 \rightarrow 1-r = 1/4 \rightarrow r = 3/4$.

You prepared. You showed up. Now go write that exam. Ghana.

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